S3 — Can the singlet superfluid explain the Hubble tension or dark matter? Numerical simulations

Supplement to both manuscripts of PROJECT spacetimes: an Einstein–Boltzmann solver written for the project, its validation, the dark radiation and the superfluid dark matter of the model, and the verdicts — with the provenance of every number

Prepared for Patrick Lee Nash, Ph.D.

November 12, 2026

0. Purpose and summary

The task states two hypotheses about the superfluid of Supplement S1 — that it can explain the observations of dark matter, and that it can explain the Hubble tension — and asks for a discussion, *in detail and using precise numerical simulations*, of why the model can or cannot do so. This supplement gives the simulations and the answers.

Everything was computed with a program written for this project in Rust (crate `spacetimes`, no external libraries): an Einstein–Boltzmann solver that follows the expansion of the universe, recombination, and the linear perturbations of all species, and computes the spectra of the microwave background and of matter. It is validated against the established code CLASS (Sec. 2). The physics of the model enters through two additions: a *dark fluid of radiation with a step* (Sec. 3) and *dark matter that is a superfluid* (Sec. 4).

Summary of the answers (each is derived below, and qualified in Sec. 5):

1. *Hubble tension.* The model contains dark radiation — the massless phase mode of the Vorton-Scalar, in equilibrium with the fermions while these are relativistic — and such radiation does raise the Hubble constant inferred from the microwave background. But for data that agree with the standard model ΛCDM, as the data of Planck and of the Atacama Cosmology Telescope do, the price is too high. In our simulation of Planck-quality data, raising H_0 to the locally measured value of 73.5 km s^{-1} Mpc^{-1} costs $\sqrt{\Delta\chi^2} = 4.0$ in the most favourable case (the fluid of the model with its step complete before the epochs that the microwave background probes), 4.3 for a fluid without step, and 8.5 to 14.5 if the step falls into the epochs probed; at a cost of $\Delta\chi^2 = 4$ the gain is at most 3.0 km s^{-1} Mpc^{-1} of the 6.1 needed. **The model cannot explain the Hubble tension.**
2. *Dark matter.* Treated as a fluid with the sound speed of the superfluid, the condensate suppresses the growth of small-scale structure. Requiring no more suppression than the Lyman-α forest allows gives, in the weak-pairing regime, a fermion mass above 1.51 keV — at which the fermions are not degenerate, hence not superfluid, in any observed halo — and, for a condensate of bound pairs, a sound speed so small that the pressure of the condensate can support cores no larger than 0.37 pc. **The superfluid state cannot be what shapes galaxies.** The *particles* of the model — bound spin-0 pairs or larger clusters with masses above roughly 0.2 GeV — remain a possible form of cold dark matter, not distinguishable in these tests from any other.

0.1 How to reproduce

```
cd "/Users/nsh/Developer/github/spacetimes"
S="cargo run --release --manifest-path code/spacetimes-rs/Cargo.toml --"
$S background      # expansion history, recombination; data/step_thermo.csv
$S spectrum        # spectra of the fiducial model, convergence test; data/cmb_*.csv
python3 code/scripts/validate_cmb_against_class.py     # comparison with CLASS
$S forecast        # fits of dark-radiation models to mock data; data/dr_*.csv (about 40 min)
$S superfluid      # superfluid dark matter; data/sfdm_*.csv (about 10 min)
$S halos           # degeneracy, condensation, self-interaction; data/dm_*.csv
python3 code/scripts/make_figures.py && python3 code/scripts/make_tables.py
```

---

1. What the model contains, cosmologically

The four-dimensional model of Supplement S1 has, besides the Standard Model, a *dark sector*:

Constituent	Degrees of freedom	Mass
fermions ψ (field dirac1671 or dirac1644)	16 (eight internal states each of particles and antiparticles)	$m = y_{VS} v_{VS}/\sqrt{2}$
radial mode s_{VS}	1	$m_s = \sqrt{2\lambda_{VS}}\, v_{VS}$
phase mode ϑ_{VS}	1	0

They interact with each other with the strength $g_{VS} = m/v_{VS}$, and with ordinary matter only through gravity (any further coupling would be an addition to the model). A dark sector that was populated in the early universe has a temperature of its own, T_d. Its history depends on the fermion mass, and two regimes must be distinguished:

- **Light fermions**, m between about 1 and 100 eV. The fermions are relativistic until late. As long as $T_d \gg m$ the sector is a tightly coupled gas of radiation (the Yukawa interaction keeps it in equilibrium; it does not stream freely). When T_d falls below m, the fermions and antifermions annihilate into phase quanta; entropy is conserved, so the phase quanta are heated, and the energy density of the dark radiation rises relative to that of radiation that simply redshifts. This **step** is the mechanism of the "stepped dark radiation" proposed for the Hubble tension by Aloni, Berlin, Joseph, Schmaltz and Weiner. If the radial mode is heavier than the fermions, the gas has 1 bosonic and 16 fermionic degrees of freedom before the step and 1 bosonic after it, and the step factor is

$$r_{\mathrm{step}} = \left[\frac{1 + \frac{7}{8}\cdot 16}{1}\right]^{1/3} = 15^{1/3} = 2.466,$$

  much larger than the $(15/7)^{1/3} = 1.289$ of the Wess–Zumino model of those authors.
- **Heavy fermions**, $m \gtrsim$ keV. The annihilation happens long before the epochs that the microwave background probes. What remains is (i) the phase quanta, a small amount of radiation, and (ii) if there is an excess of fermions over antifermions, the excess fermions, which are non-relativistic matter and pair up as described in S1: a candidate for dark matter.

A single field has a single mass, so **the model cannot be in both regimes at once**: with eV-scale fermions it has interesting dark radiation and no dark matter (fermions that light are excluded as dark matter, Sec. 4); with heavy fermions it has a dark-matter candidate and almost no dark radiation. The two hypotheses of the task are therefore tested separately, in Secs. 3 and 4.

---

2. The Einstein–Boltzmann solver and its validation

2.1 What is computed

The program follows the standard treatment of linear cosmological perturbations in the synchronous gauge (Ma and Bertschinger): photons with polarization, massless neutrinos, baryons, cold dark matter, a cosmological constant, and the dark fluid.

- *Background* (`cosmo.rs`): the Hubble rate from the densities; conformal time by Gauss–Legendre quadrature.
- *Recombination* (`cosmo.rs`): Saha equilibrium followed by the rate equations of the effective three-level atoms of hydrogen and helium in the form of the program RECFAST 1.5.2 (Seager, Sasselov and Scott; Wong, Moss and Scott), including its corrections that reproduce the multi-level calculations; reionization as a step whose redshift is fitted to the given optical depth; the helium abundance from standard nucleosynthesis, including the effect of extra radiation.
- *Perturbations* (`boltzmann.rs`): 50 coupled equations per wavenumber, integrated with an adaptive Runge–Kutta method (Dormand–Prince); the tight-coupling approximation of first order at early times; the radiation-streaming approximation at late times for the matter spectrum at large wavenumbers.
- *Spectra* (`spectra.rs`, `bessel.rs`): line-of-sight integration (Seljak and Zaldarriaga) with tabulated spherical Bessel functions; temperature (TT), polarization (EE) and cross (TE) spectra to multipole 2500; the matter power spectrum and σ_8, the rms density contrast in spheres of $8\,h^{-1}$ Mpc.

Gravitational lensing of the microwave background is not computed (see Sec. 2.4). One model takes four seconds on two processor cores.

The **fiducial model** throughout is the ΛCDM model of the author's NCSS project: $\omega_b = 0.02237$, $\omega_c = 0.1200$ (the densities of baryons and of cold dark matter in units of the critical density for $h = 1$), $h = 0.6736$ ($H_0 = 100\,h$ km s^{-1} Mpc^{-1}), $A_s = 2.1 \times 10^{-9}$, $n_s = 0.9649$ (amplitude and tilt of the primordial spectrum), $\tau = 0.0544$ (optical depth), three massless neutrino species with $N_{\mathrm{eff}} = 3.044$.

2.2 Comparison with CLASS

The reference is a set of spectra of the fiducial model computed with CLASS v3.4.2 for the NCSS project (`data/reference/`, copied unchanged, with checksums). Those spectra include lensing. To compare like with like, the script `validate_cmb_against_class.py` lenses the Rust spectra with the lensing potential *of the reference*, by the flat-sky correlation-function method of Lewis and Challinor (formulas in the header of the script; the rms deflection angle obtained from the reference potential is 2.47 arcmin, the known value). An independent calculation of the lensing correction to first order in harmonic space agrees with the script within the size of the neglected terms.

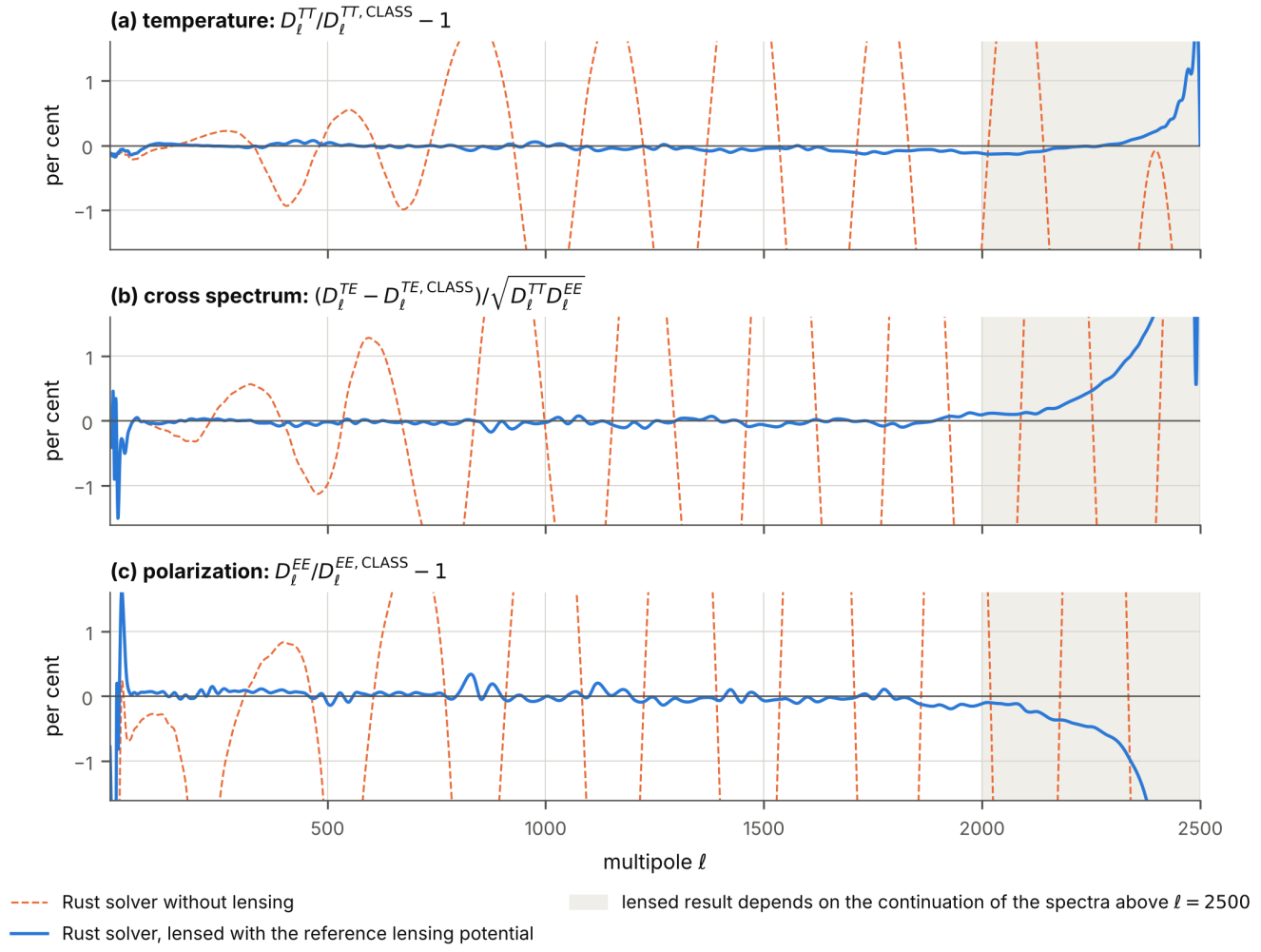


Figure 1: Differences between the Rust solver and CLASS for the fiducial model. Dashed: Rust spectra without lensing; the oscillating difference is the smoothing of the acoustic peaks by lensing, which CLASS includes. Solid: Rust spectra lensed with the lensing potential of the reference. Data: `data/cmb_validation.csv`.

multipoles	TT max	TT rms	TE max	TE rms	EE max	EE rms
2–29	0.18	0.13	1.50	0.67	7.00	2.78
30–99	0.15	0.09	0.50	0.18	1.60	0.41
100–499	0.07	0.03	0.09	0.04	0.15	0.07
500–999	0.08	0.03	0.17	0.05	0.34	0.09
1000–1499	0.11	0.06	0.11	0.05	0.20	0.07
1500–2000	0.13	0.08	0.12	0.06	0.19	0.09
2001–2500	2.00	0.41	13.40	1.53	8.65	2.31

Differences in per cent between the Rust spectra, lensed with the lensing potential of the reference, and CLASS: $D_\ell/D_\ell^{\mathrm{CLASS}} - 1$ for TT and EE, $(D_\ell^{TE} - D_\ell^{TE,\mathrm{CLASS}})/\sqrt{D_\ell^{TT} D_\ell^{EE}}$ for TE. Below $\ell = 30$ the polarization depends on the description of reionization; above $\ell = 2000$ the lensed result depends on the continuation of the spectra beyond $\ell = 2500$. Source: `data/cmb_validation.csv`.

quantity	Rust solver	CLASS v3.4.2
$100\,\theta_s$	1.040358	1.040374
r_{drag} (Mpc)	147.1099	147.1142
σ_8	0.82275	0.82291
z_*	1089.915	—
z_{drag}	1059.930	—
z_{reio}	7.6875	—

Source: header of `data/cmb_fiducial_cl.csv`; CLASS values from the header of the reference file.

For $100 \le \ell \le 2000$ the two codes agree within 0.13 per cent in TT, 0.17 per cent of $\sqrt{D^{TT}D^{EE}}$ in TE and 0.34 per cent in EE; the matter power spectrum agrees within 0.44 per cent for $10^{-4} \le k \le 3$ Mpc^{-1} (`data/pk_validation.csv`). The angular size of the sound horizon, the quantity that the microwave background measures best, agrees to 1.3×10^{-5}.

This agreement was not obtained at once. The comparison exposed five errors of the first version of the program — a wrong restriction of the wavenumber integral, an inconsistent approximation for the matter spectrum, an outdated recombination model, the loss of singly ionized helium through a numerical fault, and an inaccurate interpolation in the comparison itself. They are recorded as E15 to E20 in the error log (`provenance/P9_error_log`), with the unit tests that now guard against them.

2.3 Numerical convergence

Halving all tolerances and all wavenumber steps changes the spectra by less than 3.4×10^{-4} for $\ell \ge 100$ (`data/cmb_convergence.csv`; the command `spectrum` fails if the limit of 5×10^{-4} is exceeded). Between $\ell = 30$ and 100 the change is up to 0.1 per cent in TT and 0.6 per cent in EE, far below the cosmic variance of a single multipole there (10 to 20 per cent).

2.4 Limits of validity

1. *No lensing in the mock analyses.* The mock data of Sec. 3 and the models fitted to them are both unlensed. Lensing smooths the acoustic peaks slightly and thereby removes a little of the information on extra radiation; the forecast errors below are therefore somewhat optimistic. The comparison with published analyses of real data (Sec. 3.6) shows by how much.
2. *Massless neutrinos.* The fiducial model has no neutrino mass.
3. *Linear theory.* All statements about the matter spectrum are linear.
4. *One fluid slot.* The dark fluid is either dark radiation or superfluid dark matter; the two are not combined (they belong to different regimes, Sec. 1).

---

3. The Hubble tension

3.1 The mechanism, and the measure of success

The microwave background measures the angle $\theta_s = r_s/D_A$ under which the sound horizon r_s at recombination is seen at the distance D_A. In ΛCDM, θ_s and the densities fix $H_0 = 67.4$; the local distance network gives 73.50 ± 0.81 (H0DN collaboration). Extra radiation before recombination speeds up the expansion and shrinks r_s; to keep θ_s, D_A must shrink too, which means a larger H_0. The difficulty is that extra radiation also changes the damping of the small-scale anisotropies and the phases of the acoustic peaks, which the data constrain.

The measure of success used here is therefore: **how much does the fit to the microwave background worsen when the model is forced to give a certain H_0?**

3.2 The dark radiation of the model

The dark fluid is described by its density today, N, in units of the density of one massless neutrino species (the convention of "N_{eff}"), and by the transition redshift z_t, defined by $m = T_{d0}(1+z_t)$ with T_{d0} the temperature of the dark radiation today. Its equation of state and sound speed follow from the thermodynamics of the mixture of one massless boson and sixteen fermionic degrees of freedom of mass m at constant entropy (`cosmo.rs`, `StepTable`; unit tests check the identity $\rho = 3P - x\,dP/dx$ with $x = m/T$, the limits, and r_{step}). The fluid is tightly coupled: it has density and velocity perturbations but no anisotropic stress, in contrast to free-streaming neutrinos.

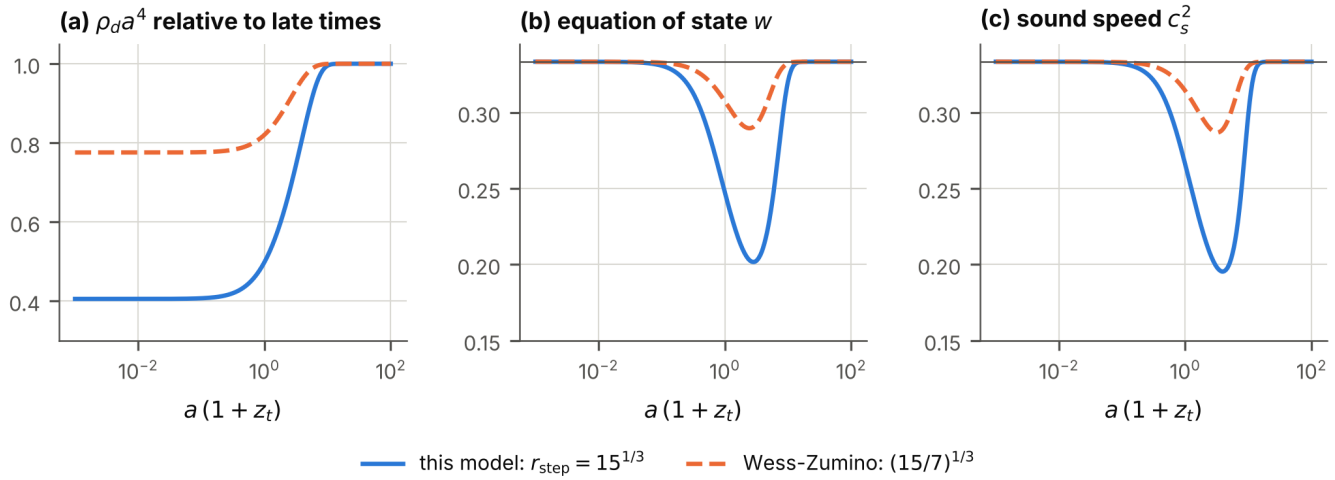


Figure 2: The stepped dark fluid of the model (solid) and, for comparison, the Wess–Zumino fluid of Aloni et al. (dashed). (a) Energy density relative to radiation that only redshifts, normalized to late times: before the step the density is lower by $1/r_{\mathrm{step}}$. (b), (c) During the step the pressure and the sound speed drop below the values 1/3 of pure radiation. Data: `data/step_thermo.csv`.

Seven one-parameter families of models are compared; in each, N is varied:

No.	Family	Dark radiation
0	free-streaming	additional massless neutrinos (the textbook ΔN_{eff})
1	fluid, no step	the fluid with $z_t \to \infty$: phase quanta of a model with heavy fermions, if they interact enough to be a fluid
2, 3, 4	fluid of the model, step in the epochs probed	$r_{\mathrm{step}} = 15^{1/3}$, $z_t = 10^4$, 2×10^4, 5×10^4
5	Wess–Zumino fluid	$r_{\mathrm{step}} = (15/7)^{1/3}$, $z_t = 2\times10^4$ (the best-fit region of Aloni et al.)
6	fluid of the model, early step	$r_{\mathrm{step}} = 15^{1/3}$, $z_t = 10^6$: after nucleosynthesis, before the scales of the microwave background enter the horizon

For $N = 0.4$ the transition redshifts 10^4, $10^{5.5}$ and 10^6 correspond to the fermion masses 1.5, 48 and 150 eV (`make_tables_cosmo.py`, from $m = T_{d0}(1+z_t)$).

3.3 Mock data and method

Mock data. The "observed" spectra are those of the fiducial ΛCDM model, without a noise realization, so that ΛCDM itself fits with $\chi^2 = 0$ and every χ^2 below is a $\Delta\chi^2$ relative to ΛCDM. Two data sets are simulated (`forecast.rs`):

- *Planck-like*: 60 per cent of the sky; the noise and beams of the three Planck channels at 100, 143 and 217 GHz; $30 \leq \ell \leq 2500$ in temperature and $30 \leq \ell \leq 2000$ in polarization; a prior of width 0.0073 on the optical depth in place of the large-scale polarization.
- *Planck-like plus deep survey*: in addition, 25 per cent of the sky observed with a beam of 1.4 arcmin and a noise of 10 µK arcmin, the sensitivity of the sixth data release of the Atacama Cosmology Telescope.

Likelihood. The exact likelihood of Gaussian fields on a fraction of the sky (formula in the header of `forecast.rs`); its gradient and curvature are tested against finite differences (unit test `forecast::tests::gradient_and_curvature_against_finite_differences`).

Fits. For each family and each N the six ΛCDM parameters are re-fitted. A fit alternates exact computations of the spectra with Newton minimizations of the exact likelihood of spectra linearized with stored derivatives; the reported χ^2 is always that of an exact computation. The stored derivatives are those of the fiducial point; recomputing them at the end point of a fit at $N = 0.5$ and continuing changes χ^2 by at most 0.053 and H_0 by at most 0.052 km s^{-1} Mpc^{-1} (`data/dr_chord_check.csv`), which bounds the error of the procedure.

Does the mock resemble the real experiment? The table compares the errors of the ΛCDM parameters predicted for the Planck-like mock with those published by the Planck collaboration:

parameter	mock, Planck-like	Planck 2018 (TT,TE,EE+lowE)	mock, with deep survey
ω_b	1.3×10^{-4}	1.5×10^{-4}	6.8×10^{-5}
ω_c	1.3×10^{-3}	1.4×10^{-3}	9.2×10^{-4}
h	5.6×10^{-3}	6.0×10^{-3}	3.7×10^{-3}
$\ln(10^{10} A_s)$	1.5×10^{-2}	1.6×10^{-2}	1.5×10^{-2}
n_s	3.3×10^{-3}	4.4×10^{-3}	2.5×10^{-3}
τ	7.2×10^{-3}	7.6×10^{-3}	7.2×10^{-3}

One-standard-deviation errors of the six ΛCDM parameters from the Fisher matrix of the mock data, compared with the published errors of the Planck collaboration (for h: the error of H_0 divided by 100). Source: `data/dr_fisher.csv`.

The agreement is within 25 per cent for every parameter. For free-streaming radiation the mock gives $\sigma(N) = 0.143$, the Planck collaboration found $N_{\mathrm{eff}} = 2.92 \pm 0.19$: the mock is more constraining by a quarter, as expected from Sec. 2.4. All $\Delta\chi^2$ below should be read with this in mind: for the real Planck data they would be smaller by a factor of about $(0.14/0.19)^2 \approx 0.6$.

3.4 Results

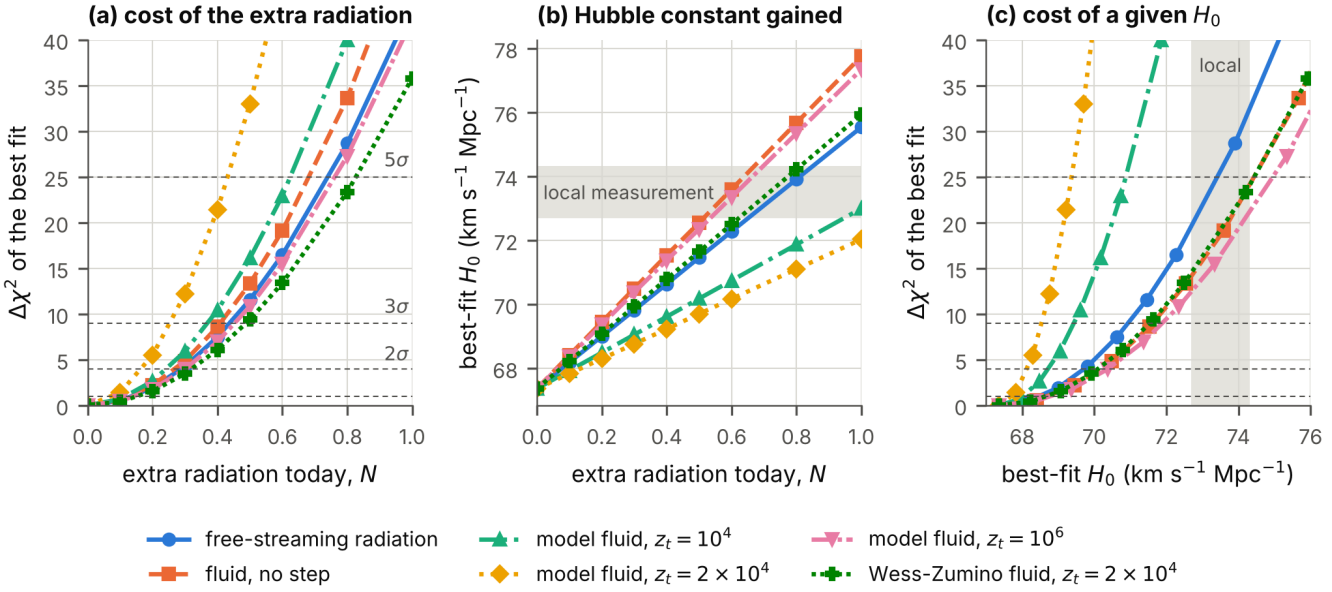


Figure 3: Planck-like mock data. (a) Worsening of the best fit, relative to ΛCDM, as a function of the density N of the extra radiation, for six of the seven families (the family with $z_t = 5 \times 10^4$ is omitted from the figure for clarity; it is in the tables). (b) The Hubble constant of the best fit; the band is the local measurement. (c) The same information combined: the cost of a given Hubble constant. Data: `data/dr_profiles.csv`.

family	$\sigma(N)$	dH_0/dN	N at $\Delta\chi^2=4$	H_0 there	H_0 at $\Delta\chi^2=9$	$N_{73.5}$	its $\Delta\chi^2$	$\sqrt{\Delta\chi^2}$
free-streaming radiation	0.143	8.19	0.29	69.7	71.0	0.75	25	5.0
fluid, no step	0.135	10.44	0.27	70.2	71.6	0.59	19	4.3
model fluid, $z_t = 10^4$	0.122	5.63	0.25	68.7	69.5	1.09*	72*	8.5*
model fluid, $z_t = 2 \times 10^4$	0.085	4.68	0.17	68.2	68.6	1.32*	211*	14.5*
model fluid, $z_t = 5 \times 10^4$	0.060	9.28	0.12	68.5	69.1	0.66	113	10.6
Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.160	8.58	0.32	70.1	71.6	0.72	19	4.3
model fluid, $z_t = 10^6$	0.150	10.04	0.30	70.4	71.9	0.62	16	4.0

Mock data set: Planck-like. $\sigma(N)$ and dH_0/dN (km s^{-1} Mpc^{-1} per unit N) from the Fisher matrix at the fiducial point; the other columns from the exact fits (`data/dr_profiles.csv`), interpolated as described in `code/scripts/make_tables_cosmo.py`. $N_{73.5}$ is the density at which the best-fit H_0 equals the local value 73.5. An asterisk marks a linear extrapolation beyond the largest fitted N.

family	N	$\Delta\chi^2$	H_0	ω_c	n_s	σ_8	S_8	r_{drag}	ΔN_{BBN}
free-streaming radiation	0.2	1.92	69.00	0.1223	0.9750	0.8288	0.835	145.26	0.200
	0.5	11.55	71.45	0.1257	0.9898	0.8378	0.825	142.62	0.500
	1.0	43.83	75.53	0.1313	1.0132	0.8523	0.811	138.51	1.000
fluid, no step	0.2	2.17	69.44	0.1219	0.9724	0.8220	0.822	145.25	0.200

family	N	$\Delta\chi^2$	H_0	ω_c	n_s	σ_8	S_8	$r_{\rm drag}$	$\Delta N_{\rm BBN}$
	0.5	13.32	72.55	0.1248	0.9836	0.8210	0.795	142.59	0.500
	1.0	51.94	77.75	0.1294	1.0019	0.8192	0.754	138.44	1.000
model fluid, $z_t = 10^4$	0.2	2.64	68.49	0.1214	0.9721	0.8274	0.836	146.12	0.081
	0.5	16.08	70.17	0.1235	0.9829	0.8342	0.829	144.66	0.203
	1.0	61.23	72.99	0.1270	1.0003	0.8451	0.816	142.33	0.405
model fluid, $z_t = 2 \times 10^4$	0.2	5.47	68.29	0.1228	0.9754	0.8355	0.851	145.68	0.081
	0.5	32.98	69.70	0.1269	0.9909	0.8542	0.864	143.60	0.203
	1.0	124.24	72.02	0.1338	1.0157	0.8846	0.886	140.33	0.405
model fluid, $z_t = 5 \times 10^4$	0.2	10.81	69.22	0.1223	0.9848	0.8340	0.838	145.27	0.081
	0.5	65.29	71.97	0.1258	1.0138	0.8513	0.834	142.63	0.203
	1.0	247.48	76.63	0.1315	1.0593	0.8790	0.826	138.51	0.405
Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.2	1.55	69.07	0.1223	0.9739	0.8270	0.832	145.34	0.155
	0.5	9.38	71.64	0.1258	0.9872	0.8332	0.819	142.80	0.388
	1.0	35.77	75.93	0.1314	1.0084	0.8427	0.797	138.85	0.776
model fluid, $z_t = 10^6$	0.2	1.76	69.36	0.1220	0.9713	0.8219	0.823	145.27	0.081
	0.5	10.80	72.35	0.1250	0.9808	0.8209	0.797	142.63	0.203
	1.0	42.16	77.33	0.1299	0.9966	0.8192	0.758	138.52	0.405

Best fits to the Planck-like mock data at fixed N (fiducial ΛCDM: $H_0 = 67.36$, $\omega_c = 0.1200$, $n_s = 0.9649$, $\sigma_8 = 0.8228$, $S_8 = 0.832$, $r_{\rm drag} = 147.11$ Mpc). $\Delta N_{\rm BBN}$ is the extra radiation during nucleosynthesis that the helium abundance was computed with. Source: `data/dr_profiles.csv`.

Reading the table and Fig. 3:

1. **Every family can reach the local H_0, and every one at a prohibitive price.** The cheapest is the fluid of the model with an early step, at $\sqrt{\Delta\chi^2} = 4.0$; the fluid without step and the Wess–Zumino fluid need 4.3 and 4.3, free-streaming radiation 5.0.
2. **A step of the model that falls into the epochs probed is the most expensive case.** Its large step leaves a strong imprint: modes that enter the horizon before the step see less radiation than modes that enter after it, which changes the spectrum as a function of ℓ in a way that the other parameters cannot absorb. For z_t between 10^4 and 5×10^4 the local value costs $\sqrt{\Delta\chi^2} = 8.5$ to 14.5.
3. **An early step is slightly cheaper than no step, because of helium.** If the step is complete before the scales of the microwave background enter the horizon, these scales see a fluid of constant density N; but during nucleosynthesis the fluid had only the density $N/r_{\rm step}$, and less helium was made: at $N = 0.5$ the helium mass fraction is 0.2485 instead of 0.2523. Less helium means more free electrons and less damping of the small-scale anisotropies, which offsets part of the additional damping caused by the radiation; $\Delta\chi^2$ falls from 13.3 to 10.8.
4. **At a cost of $\Delta\chi^2 = 4$ the gain is small**: $H_0 = 70.4$ for the early step, 70.2 for the fluid without step, 69.7 for free-streaming radiation, and between 68.2 and 68.7 for a step in the epochs probed.
5. The fits compensate the extra radiation mainly by more cold dark matter and a bluer primordial spectrum (larger n_s), the well-known degeneracy; the amplitude σ_8 rises for free-streaming radiation and for the stepped fluids, and stays nearly constant for the fluid without step.

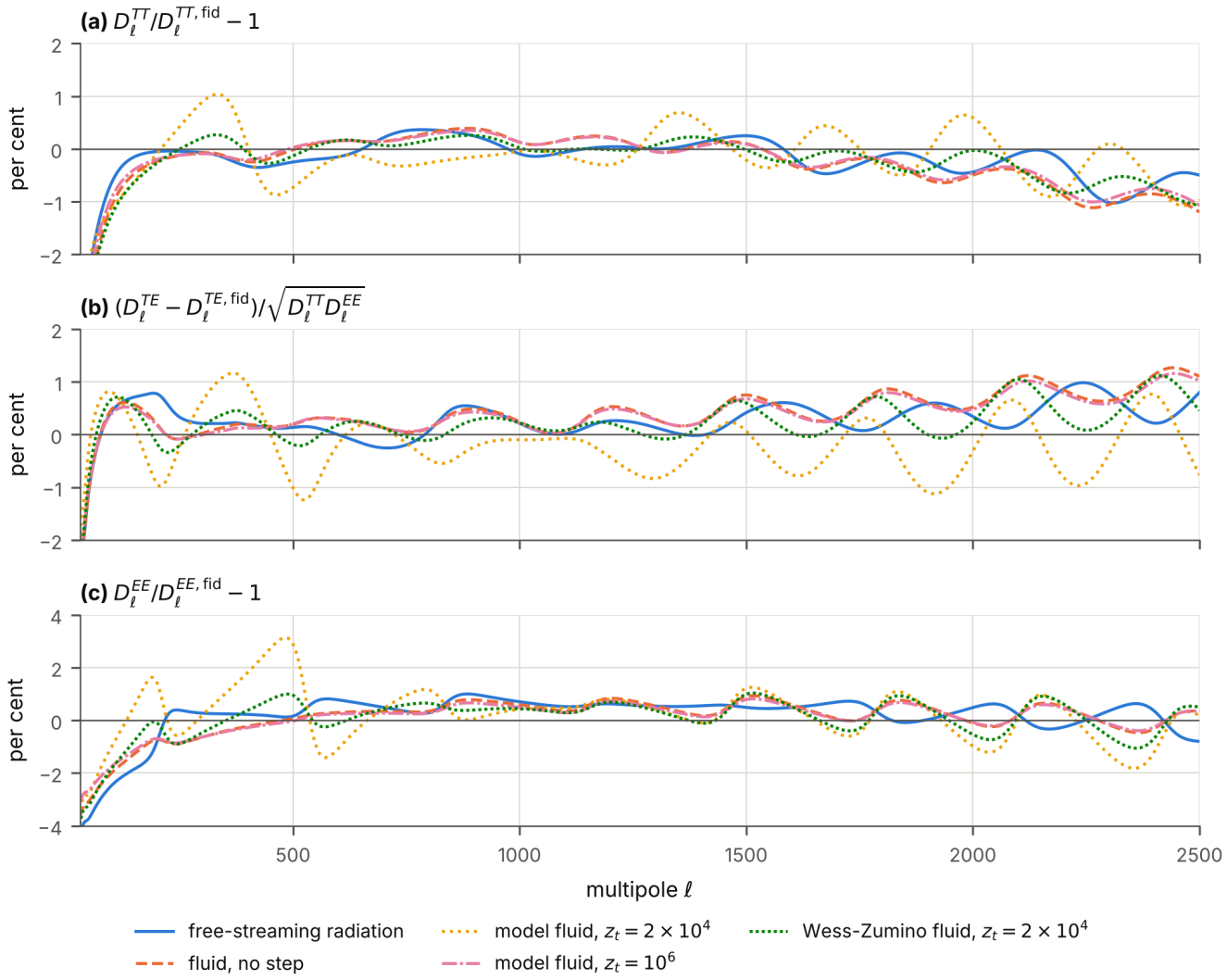


Figure 4: Residuals of the best fits at $N = 0.5$ (Planck-like mock) relative to the fiducial spectra. The fits absorb most of the effect of the extra radiation; what remains is the signal that the likelihood penalizes. The fluid of the model with its step at $z_t = 2 \times 10^4$ leaves the largest residuals (its curves leave the frame below $\ell \approx 100$, as all curves do, where the errors of the data are largest). Data: `data/dr_residuals.csv`.

Dependence on the transition redshift. For the fluid of the model with $N = 0.4$ after the step:

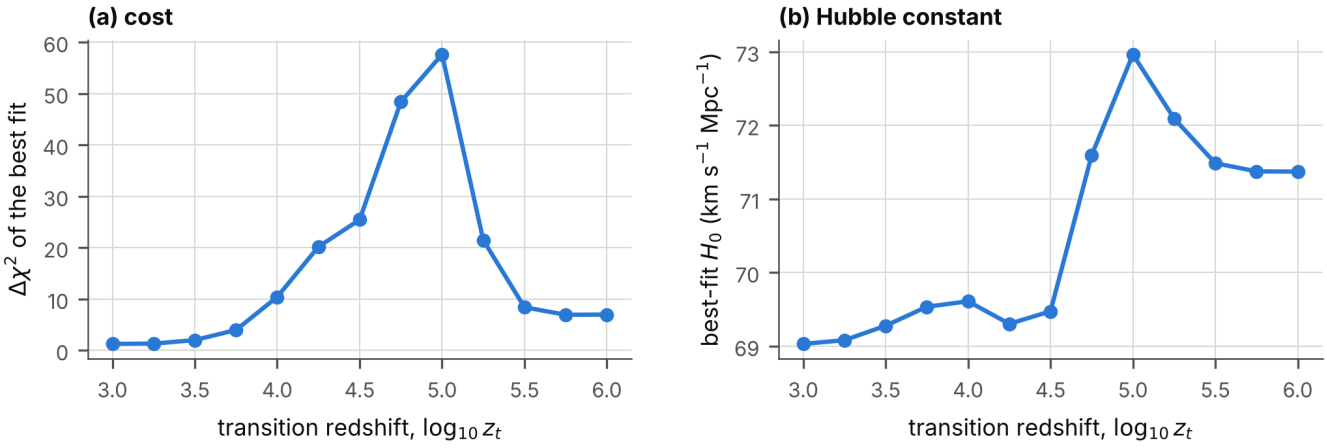


Figure 5: Fluid of the model with $N = 0.4$: cost and Hubble constant as functions of the transition redshift. A step long before the epochs probed by the microwave background ($z_t \gtrsim 10^{5.5}$) acts like a fluid without step that was less dense during nucleosynthesis ($\Delta\chi^2 = 7.0$, $H_0 = 71.4$); a step after recombination ($z_t \lesssim 10^3$) means that only N/r_{step} was present before recombination, so both the cost and the gain are small ($\Delta\chi^2 = 1.2$, $H_0 = 69.0$); in between, the step is seen by the data and is expensive. Data: `data/dr_zt_scan.csv`.

$\log_{10} z_t$	$\Delta\chi^2$	H_0	n_s	σ_8
3.00	1.24	69.03	0.9702	0.8206
3.25	1.30	69.08	0.9702	0.8208
3.50	1.99	69.28	0.9718	0.8217
3.75	3.95	69.53	0.9749	0.8247
4.00	10.39	69.61	0.9793	0.8320
4.25	20.16	69.30	0.9846	0.8450
4.50	25.43	69.47	0.9932	0.8543
4.75	48.38	71.60	1.0061	0.8407
5.00	57.61	72.96	1.0012	0.8218
5.25	21.45	72.09	0.9849	0.8198
5.50	8.39	71.49	0.9784	0.8212
5.75	6.92	71.37	0.9776	0.8213
6.00	6.96	71.37	0.9777	0.8211

Model fluid with $N = 0.4$ after the step, Planck-like mock data. Source: `data/dr_zt_scan.csv`.

With the deep survey. The second mock data set tightens everything:

family	$\sigma(N)$	dH_0/dN	N at $\Delta\chi^2 = 4$	H_0 there	H_0 at $\Delta\chi^2 = 9$	$N_{73.5}$	its $\Delta\chi^2$	$\sqrt{\Delta\chi^2}$
free-streaming radiation	0.077	7.92	0.16	68.6	69.2	0.78	95	9.7
fluid, no step	0.067	9.80	0.13	68.7	69.3	0.64	87	9.3
model fluid, $z_t = 10^4$	0.086	6.28	0.17	68.4	69.0	0.98	120	10.9
model fluid, $z_t = 2 \times 10^4$	0.059	5.40	0.12	68.0	68.3	1.15*	330*	18.2*
model fluid, $z_t = 5 \times 10^4$	0.042	6.93	0.08	67.9	68.2	0.90	412	20.3

family	$\sigma(N)$	dH_0/dN	N at $\Delta\chi^2=4$	H_0 there	H_0 at $\Delta\chi^2=9$	$N_{73.5}$	its $\Delta\chi^2$	$\sqrt{\Delta\chi^2}$
Wess–Zumino fluid, $z_t = 2\times10^4$	0.088	8.25	0.18	68.8	69.5	0.75	69	8.3
model fluid, $z_t = 10^6$	0.074	9.45	0.15	68.8	69.5	0.66	76	8.7

Mock data set: Planck-like + deep survey. $\sigma(N)$ and dH_0/dN (km s^{-1} Mpc^{-1} per unit N) from the Fisher matrix at the fiducial point; the other columns from the exact fits (`data/dr_profiles.csv`), interpolated as described in `code/scripts/make_tables_cosmo.py`. $N_{73.5}$ is the density at which the best-fit H_0 equals the local value 73.5. An asterisk marks a linear extrapolation beyond the largest fitted N.

3.5 The phase mode as a “fractional neutrino”

Even with heavy fermions the model has one unavoidable piece of dark radiation: the massless phase mode. If the dark sector was once in thermal contact with ordinary matter and decoupled when the plasma had g_* effective degrees of freedom, a massless boson contributes today (Weinberg)

$$\Delta N_{\rm eff} = \frac{4}{7}\Big(\frac{43}{4\,g_*}\Big)^{4/3} = \begin{cases} 0.027, & g_* = 106.75 \ \text{(decoupling above the electroweak scale)},\\ 0.39, & g_* = 57/4 \ \text{(just before the muons annihilate)}. \end{cases}$$

By Sec. 3.4 such an amount raises H_0 by at most $0.39\times\ 8.2 = 3.2$ km s^{-1} Mpc^{-1} — a welcome but insufficient shift, and the largest value is itself disfavoured by the data quoted next.

3.6 Comparison with published analyses of real data

The simulation above assumes that the sky is ΛCDM-like and asks what the model then costs. Whether the real data prefer extra radiation is an empirical question, which has been answered by others:

- Planck 2018: $N_{\rm eff} = 2.92 \pm 0.19$ for free-streaming radiation; no preference.
- Atacama Cosmology Telescope, sixth data release, combined with Planck (Calabrese et al.): $N_{\rm eff} = 2.86 \pm 0.13$ for free-streaming radiation, and $N_{\rm idr} < 0.134$ (95 per cent confidence) for self-interacting dark radiation — the family 1 of this supplement. With the slope of the table of Sec. 3.4 that limit allows a gain of $0.134\times\ 10.4 = 1.4$ km s^{-1} Mpc^{-1}.
- For the stepped fluid of Aloni et al. and a second interacting model, Cvetko, Joseph and Marques-Tavares find that the data of the Atacama Cosmology Telescope restrict the density about as strongly as for simple models, and conclude that they “significantly” restrict interacting dark radiation as a solution of the tension.
- Garny, Niedermann and Sloth show that the limits loosen to $\Delta N_{\rm eff} < 0.58$ if the primordial spectrum is allowed a running tilt and a running of the running — a caveat that applies equally to our fixed power-law spectrum.

Our mock results agree with this picture: for the fluid without step the forecast limits at $\Delta\chi^2 = 4$ are $N < 0.27$ from the Planck-like mock and $N < 0.13$ from the mock with the deep survey, to be compared with the published limit of 0.134 from the real data of both experiments.

3.7 Verdict on the Hubble tension

The dark radiation of the model raises H_0, and its stepped form is exactly the kind of fluid that was proposed for this purpose. But (i) for data consistent with ΛCDM, the density needed for the local value of H_0 is excluded at $\sqrt{\Delta\chi^2} \geq 4.0$ in every family simulated, and at $\sqrt{\Delta\chi^2} \geq 8.5$ if the step of the model falls into the epochs probed; (ii) the published analyses of the real data find no

preference for such radiation and limits near $N \approx 0.13$; (iii) the step of the model, $15^{1/3}$, is larger than that of the Wess–Zumino model and is *more* visible to the data, not less — unless it is complete before $z \approx 3 \times 10^5$, and then the model is the fluid of family 6, which still fails by 4.0 standard deviations. **The hypothesis that the model explains the Hubble tension is not supported: with the amount of fluid dark radiation that the published limit allows, it removes about 23 per cent of the discrepancy.** This agrees with the conclusion of the NCSS project for bound neutrino fields, now reached for a different dark sector with an independent code.

---

4. Dark matter

4.1 The superfluid as a cosmological fluid

Suppose the dark matter consists of the fermions of the model (an excess of particles over antiparticles; the size of the excess is a free number and is chosen to give the observed density). At zero temperature they are in the paired state of S1. Its response to slow, long-wavelength perturbations is that of a fluid with pressure $P(\rho)$: density perturbations propagate as first sound with $c_s^2 = dP/d\rho$ (S1, Sec. 6.4). Two limits bracket the possibilities:

- **Weak pairing** (overlapping pairs). The pressure is the degeneracy pressure of the fermions — pairing lowers it only by terms of relative order $(\Delta/E_F)^2$ — so $c_s = v_F/\sqrt{3}$. With eight internal states, mass m, and the mean cosmic density of dark matter, the Fermi velocity today is fixed by m alone, and scales as $1/a$ with the expansion (as $a \to 0$ the gas becomes relativistic, which the program includes exactly).
- **Bound pairs** (or larger bound clusters) that have Bose-condensed. If their mutual interaction is repulsive, the condensate has the equation of state of a self-interacting scalar field (Colpi, Shapiro and Wasserman): matter-like at low density, with the Bogoliubov sound speed $c_s^2 \propto \rho$, and radiation-like at high density. It is characterized by one number, the sound speed today at the mean density, c_{s0}; the radius of a self-gravitating core supported by its pressure (the Thomas–Fermi radius) is $R_{TF} = \pi\, c_s/\sqrt{4\pi G\rho}$, independent of the density.

In both limits the pressure is the *smallest* that the constituents can have: a thermal population of the same particles has more. The bounds derived below are therefore lower limits on what any state of these particles must satisfy.

4.2 Simulation

The cold dark matter of the Boltzmann solver is replaced by a fluid with the equation of state and sound speed of either limit (`cosmo.rs`, `SuperfluidDm`; unit tests check energy conservation, $c_s^2 = dP/d\rho$ and the limits; a test verifies that a fluid with negligible sound speed reproduces cold dark matter to 2×10^{-4}). The linear matter power spectrum today is computed for wavenumbers up to 250 Mpc^{-1} and divided by that of cold dark matter. The *half mode* is the wavenumber at which the ratio first falls to 1/4. It is compared with the half mode of thermal warm dark matter at the mass limits from the Lyman-α forest, 5.3 keV (Iršič et al.) and, more conservatively, 3.5 keV, using the fitting formula of Viel et al.

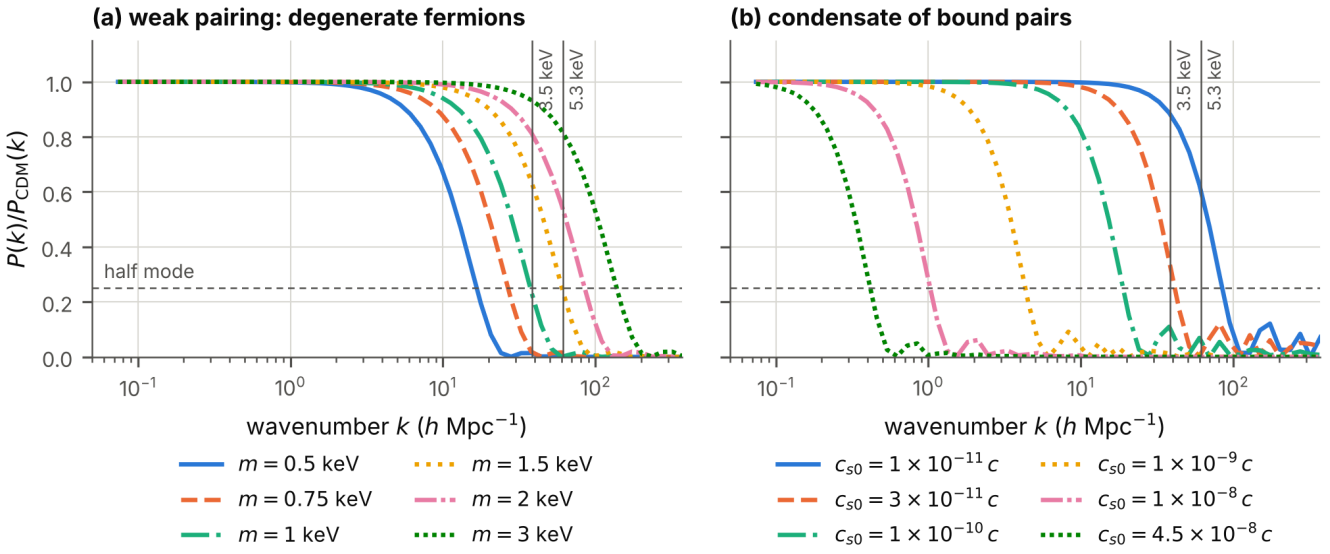


Figure 6: Linear matter power spectrum relative to cold dark matter if the dark matter is the superfluid. (a) Weak pairing, for several fermion masses. (b) Condensate of bound bosons, for several sound speeds c_{s0} today at the mean density; the largest corresponds to a Thomas–Fermi radius of 1 kpc. The vertical lines mark the half modes of thermal relics of 3.5 and 5.3 keV. The oscillations are sound waves of the dark fluid. Data: `data/sfdm_transfer.csv`.

model	c_{s0}/c	c_{s0} (m/s)	half mode (h/Mpc)	thermal relic (keV)	R_{TF}	ΔN at $a = 10^{-9}$
degenerate fermions, $m = 0.3$ keV	1.2×10^{-7}	35.8	9.2	1.0	—	3.3×10^{-3}
degenerate fermions, $m = 0.5$ keV	6.0×10^{-8}	18.1	16.7	1.6	—	1.7×10^{-3}
degenerate fermions, $m = 0.75$ keV	3.5×10^{-8}	10.6	26.7	2.5	—	9.8×10^{-4}
degenerate fermions, $m = 1$ keV	2.4×10^{-8}	7.2	37.4	3.4	—	6.7×10^{-4}
degenerate fermions, $m = 1.5$ keV	1.4×10^{-8}	4.19	60.4	5.2	—	3.9×10^{-4}
degenerate fermions, $m = 2$ keV	9.5×10^{-9}	2.86	85.0	7.1	—	2.7×10^{-4}
degenerate fermions, $m = 3$ keV	5.5×10^{-9}	1.66	137.7	11.0	—	1.6×10^{-4}
degenerate fermions, $m = 4.5$ keV	3.2×10^{-9}	0.969	223.6	17.1	—	9.2×10^{-5}
condensate, $c_{s0} = 3.0 \times 10^{-12}\ c$	3.0×10^{-12}	0.000899	185.64	14.44	0.067 pc	4.2×10^{-4}
condensate, $c_{s0} = 1.0 \times 10^{-11}\ c$	1.0×10^{-11}	0.003	84.43	7.10	0.22 pc	9.4×10^{-4}
condensate, $c_{s0} = 3.0 \times 10^{-11}\ c$	3.0×10^{-11}	0.00899	41.24	3.72	0.67 pc	1.9×10^{-3}
condensate, $c_{s0} = 1.0 \times 10^{-10}\ c$	1.0×10^{-10}	0.03	18.90	1.84	2.2 pc	4.3×10^{-3}
condensate, $c_{s0} = 1.0 \times 10^{-9}\ c$	1.0×10^{-9}	0.3	4.32	0.49	22 pc	2.0×10^{-2}
condensate, $c_{s0} = 1.0 \times 10^{-8}\ c$	1.0×10^{-8}	3	1.03	0.13	0.22 kpc	9.4×10^{-2}
condensate, $c_{s0} = 4.5 \times 10^{-8}\ c$	4.5×10^{-8}	13.5	0.42	0.06	1 kpc	2.6×10^{-1}

c_{s0}: sound speed today at the mean cosmic density. Half mode: wavenumber at which $P/P_{\rm CDM}$ first falls to 1/4. Thermal relic: mass of the thermal warm-dark-matter particle with the same half mode. R_{TF}: Thomas–Fermi radius of the condensate. Last column: density of the fluid during nucleosynthesis in units of one neutrino species. Source: `data/sfdm_summary.csv`.

benchmark: thermal relic (keV)	its half mode (h/Mpc)	fermion mass $>$ (keV)	$v_{F0} <$ (m/s)	condensate: $c_{s0}/c <$	$c_{s0} <$ (mm/s)	$R_{TF} <$ (pc)
3.5	38.5	1.02	12.07	3.3×10^{-11}	10	0.74
5.3	61.0	1.51	7.17	1.6×10^{-11}	4.9	0.37

Source: `data/sfdm_bounds.csv`.

4.3 What the bounds imply

Weak pairing. The fermion mass must exceed 1.51 keV (1.02 keV for the conservative benchmark). A paired state requires a degenerate gas, since its critical temperature is a fraction of the Fermi energy. The next table shows that fermions of such a mass are far from degenerate in every observed halo:

system	ρ ($M_\odot$/pc^3)	ρ (GeV/cm^3)	σ (km/s)	$m_{\rm deg}$ (keV)	$m_{B,\rm max}$ (eV)	v_F at 1.5 keV (km/s)	$\rho_{\rm deg}/\rho$ at 1.5 keV
dwarf spheroidal galaxy, core	0.1	3.8	10	0.276	262	1.047	8.7×10^2
Milky Way at the Sun	0.0105	0.399	156	0.020	19	0.494	3.1×10^7
cluster of galaxies, core	0.003	0.114	1000	0.004	3.45	0.325	2.9×10^{10}

$m_{\rm deg}$: fermions with eight internal states are degenerate at the density ρ and velocity dispersion σ only if they are lighter than this. $m_{B,\rm max}$: bosons are Bose-condensed only if they are lighter than this. Last two columns: Fermi velocity of 1.5 keV fermions at the density ρ, and the factor by which the density would have to be larger for them to be degenerate. Formulas in the header of `data/dm_halo_estimates.csv` (program: `spacetimes halos`).

In the densest observed dark-matter systems, the cores of dwarf spheroidal galaxies, fermions with eight internal states are degenerate only below 0.28 keV — the bound of Tremaine and Gunn for this model — whereas structure formation demands more than 1.51 keV. **In the weak-pairing regime the dark matter in galaxies is a classical gas, not a superfluid.** (The author's notebook `7+1.nb`, section 24, reached the corresponding mass window for the unpaired field with sixteen states and a thermal distribution; the present bound is the one for the coldest possible state.)

Bound pairs. The sound speed of the condensate today must be below 1.6×10^{-11} of the speed of light, and the Thomas–Fermi radius below 0.37 pc. The reason for so strong a bound is visible in Fig. 6(b): such a condensate is radiation-like in the early universe, and all perturbations that enter the horizon during that phase are erased. **The pressure of the condensate cannot produce the kiloparsec-sized cores that motivate superfluid dark matter.** For a core of 1 kpc the half mode would be at $k \approx 0.42\ h$ Mpc^{-1}, hundreds of times too small. A second, independent obstacle: the pairs can be condensed in a halo only if they are lighter than 262 eV in a dwarf galaxy (table above, column $m_{B,\rm max}$), while their self-interaction demands the opposite, as shown next.

Self-interaction. Colliding clusters of galaxies limit the cross section per mass of dark matter to about 1 cm^2/g (Markevitch et al.; Randall et al.). A pair bound by the Yukawa potential of the radial mode has a size and a scattering length of at least the range $1/m_s$, and binding requires $D_{VS} > D_c$ (S1, Eq. (5.4)). Together:

α_{VS}	smallest fermion mass (MeV)	m_s (MeV)	range $1/m_s$ (fm)	$n\lambda^3$ of the pairs in a dwarf core
0.01	4.26e+03	25.4	7.78	2.3×10^{-30}
0.1	918	54.7	3.61	1.1×10^{-27}
1	198	118	1.68	5.0×10^{-25}

α_{VS}	smallest fermion mass (MeV)	m_s (MeV)	range $1/m_s$ (fm)	$n\lambda^3$ of the pairs in a dwarf core
4	78.5	187	1.06	2.0×10^{-23}

Source: `data/dm_selfinteraction.csv` (program: `spacetimes halos`). Condensation would need $n\lambda^3 \geq 2.612$.

Bound pairs are acceptable dark matter only if the fermions are heavier than about 198 MeV (for $\alpha_{VS} = 1$), with a radial mode of nuclear range. At such masses the phase-space density in halos is smaller than the value needed for condensation by more than twenty orders of magnitude (last column).

Bound clusters. With eight internal states and an attraction between all of them, up to eight fermions can occupy the same orbital, and bound states of more than two fermions are expected, as nuclei are bound states of nucleons; such "nuggets" of fermions bound by a scalar field have been studied as dark matter (Wise and Zhang; Gresham, Lou and Zurek). Their interior is degenerate and, by S1, paired: the superfluid would exist *inside* the dark-matter particles, as it does inside nuclei and neutron stars, without any effect on galactic scales. Supplement S4 (Sec. 5) computes the limit of large clusters from the field equation of the Vorton-Scalar in matter: matter is self-bound if $C = g_{VS}^2 m^2/m_s^2$ exceeds 3.7011 (5.9218 if the energy of the negative-energy states is included), and it is then in a state in which the Vorton-Scalar has no expectation value and the fermions are massless. For the example of the table above with $\alpha_{VS} = 1$, $C = 35.5$, and a fermion inside bulk matter weighs 43 per cent less than a free one (17 per cent with the negative-energy states). Whether such clusters form, and with what abundance, is outside this project.

4.4 Verdict on dark matter

The hypothesis has two readings, and they have different answers.

- *The superfluid state explains the observed properties of dark matter* (as in proposals where a condensate of galactic size shapes rotation curves or cores). **No.** The simulations exclude both ways in which the condensate could act on galactic scales: degenerate fermions light enough to be superfluid in halos erase small-scale structure, and a condensate of bound pairs with enough pressure to support cores does so even more strongly.
- *The dark matter is made of the particles whose condensate is the superfluid.* **Possible, and untested by these observations.** Heavy bound pairs or larger clusters, with an abundance set by an unexplained excess of particles over antiparticles, behave as cold dark matter with a self-interaction of nuclear size. Nothing in the present data distinguishes them from other cold dark matter, and the model does not predict their abundance.

---

5. What has been established, and what has not

Established:

1. The Boltzmann solver agrees with CLASS to the precision stated in Sec. 2.2; its results are converged (Sec. 2.3).
2. For Planck-quality data that follow ΛCDM, the cost of raising H_0 with each of seven kinds of dark radiation, including the stepped fluid that the model predicts (Sec. 3.4), and its dependence on the transition redshift.
3. The suppression of structure by superfluid dark matter in both limits, and the resulting bounds (Sec. 4.2); the elementary conditions for degeneracy, condensation and self-interaction in halos (Sec. 4.3).

Not established, or assumed:

1. The mock analysis is not an analysis of the real data. It has no lensing, no foregrounds, an idealized likelihood, and a power-law primordial spectrum; its limits on N are tighter than those of Planck by about a quarter (Sec. 3.3). The conclusions about the real sky rest on the published analyses cited in Sec. 3.6, with which the mock results are consistent.
2. The step was simulated for $m_s \gg m$ (one boson after the step). If the radial mode is lighter than the fermions there are two smaller steps instead of one; this was not simulated.
3. The dark radiation was assumed to be a perfect fluid at all times. After the step the phase quanta interact only through derivative couplings and may begin to stream freely; family 0 and family 1 bracket this.
4. The translation of the Lyman-α limits into a half mode uses a fitting formula for thermal relics; the spectrum of a fluid with sound waves has a different shape beyond the half mode. A dedicated analysis of the Lyman-α data could shift the bounds by a factor of order one, not by the orders of magnitude that separate them from the interesting values.
5. For bound pairs, a repulsive pair–pair interaction was assumed; with eight internal states the interaction between pairs may be attractive, in which case the condensate is unstable to the formation of larger clusters (Sec. 4.3).
6. The abundance of the dark matter, the origin of the particle excess, the thermal history of the dark sector, and the formation of bound states.
7. The fermion mass was taken to be constant through the step. Supplement S4 (Sec. 6) finds that the vacuum value of the Vorton-Scalar, and with it the mass, melts at a temperature $T_{1/2}$ with $T_{1/2}/m$ between 0.4 and 7 for the couplings examined; the assumption holds if $T_{1/2} \gg m/3$, which requires $g_{VS} \lesssim 0.3$. For stronger coupling the step would coincide with that transition, which was not simulated.
8. The sound speed of the degenerate gas was that of free fermions. The correction from the mean field of the Vorton-Scalar is of relative order $k_F|a|$ (a the scattering length), which the limit on the self-interaction keeps below 6.4×10^{-13} in a dwarf galaxy for fermions of 1 keV (S4, Sec. 5.5).

---

6. Provenance of the numbers and figures

Item	Produced by	File
fiducial spectra, derived parameters	`spacetimes spectrum`	`data/cmb_fiducial_cl.csv`, `cmb_fiducial_pk.csv`
convergence	`spacetimes spectrum`	`data/cmb_convergence.csv`
CLASS reference	copied from the NCSS project (checksums in the headers)	`data/reference/class_lcdm_lensed_cl.csv`, `class_lcdm_pk.csv`
comparison with CLASS (Fig. 1, tables of Sec. 2.2)	`validate_cmb_against_class.py`	`data/cmb_validation.csv`, `pk_validation.csv`
thermodynamics of the step (Fig. 2)	`spacetimes background`	`data/step_thermo.csv`
Fisher errors	`spacetimes forecast`	`data/dr_fisher.csv`
fits at fixed N (Figs. 3, 4; tables of Sec. 3.4)	`spacetimes forecast`	`data/dr_profiles.csv`, `dr_residuals.csv`
accuracy of the fitting procedure	`spacetimes forecast`	`data/dr_chord_check.csv`
scan of the transition redshift (Fig. 5)	`spacetimes forecast`	`data/dr_zt_scan.csv`
superfluid dark matter (Fig. 6, tables of Sec. 4.2)	`spacetimes superfluid`	`data/sfdm_transfer.csv`, `sfdm_summary.csv`, `sfdm_bounds.csv`
halos, self-interaction (tables of Sec. 4.3)	`spacetimes halos`	`data/dm_halo_estimates.csv`, `dm_selfinteraction.csv`

Item	Produced by	File
interpolated quantities (N and H_0 at given $\Delta\chi^2$; cost of the local H_0)	`make_tables_cosmo.py` (method in its header)	`provenance/generated/tables.md`
figures	`make_figures.py`	`figures/`

References

1. C.-P. Ma and E. Bertschinger, *Cosmological perturbation theory in the synchronous and conformal Newtonian gauges*, Astrophys. J. **455**, 7 (1995).
2. U. Seljak and M. Zaldarriaga, *A line-of-sight integration approach to cosmic microwave background anisotropies*, Astrophys. J. **469**, 437 (1996).
3. S. Seager, D. D. Sasselov and D. Scott, *A new calculation of the recombination epoch*, Astrophys. J. **523**, L1 (1999); *How exactly did the universe become neutral?*, Astrophys. J. Suppl. **128**, 407 (2000).
4. W. Y. Wong, A. Moss and D. Scott, *How well do we understand cosmological recombination?*, Mon. Not. R. Astron. Soc. **386**, 1023 (2008).
5. D. Blas, J. Lesgourgues and T. Tram, *The Cosmic Linear Anisotropy Solving System (CLASS). Part II: Approximation schemes*, J. Cosmol. Astropart. Phys. 07 (2011) 034.
6. A. Lewis and A. Challinor, *Weak gravitational lensing of the CMB*, Phys. Rep. **429**, 1 (2006).
7. Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020).
8. H0DN Collaboration (S. Casertano et al.), *The Local Distance Network: a community consensus report on the measurement of the Hubble constant at ~1% precision*, Astron. Astrophys. (2026), doi:10.1051/0004-6361/202557993; arXiv:2510.23823.
9. D. Aloni, A. Berlin, M. Joseph, M. Schmaltz and N. Weiner, *A step in understanding the Hubble tension*, Phys. Rev. D **105**, 123516 (2022).
10. E. Calabrese et al. (Atacama Cosmology Telescope), *The Atacama Cosmology Telescope: DR6 constraints on extended cosmological models*, J. Cosmol. Astropart. Phys. 11 (2025) 063; arXiv:2503.14454.
11. W. Cvetko, M. Joseph and G. Marques-Tavares, *InterACTing dark radiation models after ACT*, arXiv:2512.19633 (2025).
12. M. Garny, F. Niedermann and M. S. Sloth, *Spectral running, interacting dark radiation, and the Hubble tension in light of ACT DR6 data*, Phys. Rev. D (accepted 13 October 2026), doi:10.1103/c68x-58mj.
13. S. Weinberg, *Goldstone bosons as fractional cosmic neutrinos*, Phys. Rev. Lett. **110**, 241301 (2013).
14. S. Tremaine and J. E. Gunn, *Dynamical role of light neutral leptons in cosmology*, Phys. Rev. Lett. **42**, 407 (1979).
15. M. Viel, J. Lesgourgues, M. G. Haehnelt, S. Matarrese and A. Riotto, *Constraining warm dark matter candidates including sterile neutrinos and light gravitinos with WMAP and the Lyman-α forest*, Phys. Rev. D **71**, 063534 (2005).
16. V. Iršič et al., *New constraints on the free-streaming of warm dark matter from intermediate and small scale Lyman-α forest data*, Phys. Rev. D **96**, 023522 (2017).
17. M. Colpi, S. L. Shapiro and I. Wasserman, *Boson stars: gravitational equilibria of self-interacting scalar fields*, Phys. Rev. Lett. **57**, 2485 (1986).
18. B. Li, T. Rindler-Daller and P. R. Shapiro, *Cosmological constraints on Bose–Einstein-condensed scalar field dark matter*, Phys. Rev. D **89**, 083536 (2014).
19. L. Berezhiani and J. Khoury, *Theory of dark matter superfluidity*, Phys. Rev. D **92**, 103510 (2015).
20. M. Markevitch et al., *Direct constraints on the dark matter self-interaction cross section from the merging galaxy cluster 1E 0657–56*, Astrophys. J. **606**, 819 (2004); S. W. Randall et al., Astrophys. J. **679**, 1173 (2008).
21. M. B. Wise and Y. Zhang, *Stable bound states of asymmetric dark matter*, Phys. Rev. D **90**,

055030 (2014); M. I. Gresham, H. K. Lou and K. M. Zurek, *Nuclear structure of bound states of asymmetric dark matter*, Phys. Rev. D **96**, 096012 (2017).
22. P. L. Nash, notebook `7+1.nb`, section 24 (see `provenance/P3_notebooks_used`); the NCSS project, github.com/once-ere/NCSS.